components of adjacent nucleotides are called **phosphodiester bond** (5'⟶3'), indicating the polarity of the strand.

The ends of the DNA or RNA are distinct. The two ends are designated by the symbols 5' and 3'. The symbol 5' refers to carbon in the sugar to which a phosphate (PO_4) functional group is attached. The symbol 3' refers to carbon in the sugar to which hydroxyl (OH) functional group is attached. In RNA, every nucleotide residue has an additional $^{-}$OH group at 2' position in the ribose. Understanding the 5'⟶3' direction of a nucleic acid is critical for understanding the aspects of replication and transcription.

Based on the X - ray diffraction analysis of **Maurice Wilkins** and **Rosalind Franklin**, the double helix model for DNA was proposed by **James Watson** and **Francis Crick in 1953**. The highlight was the base pairing between the two strands of the polynucleotide chain. This proposition was based on the observations of Erwin Chargaff that Adenine pairs with Thymine (A = T) with two hydrogen bonds and Guanine pairs with Cytosine (G ≡ C) with three hydrogen bonds. The ratios between Adenine with Thymine and Guanine with Cytosine are constant and equal. The base pairing confers a unique property to the polynucleotide chain. They are said to be complementary to each other, that is, if the sequence of bases in one strand (template) is known, then the sequence in the other strand can be predicted. The salient features of DNA structure has already been dealt in class XI.

5.5 RNA world

A typical cell contains about ten times as much RNA as DNA. The high RNA content is mainly due to the variety of roles played by RNA in the cell. **Fraenkel-Conrat** and **Singer** (1957) first demonstrated that **RNA** is the genetic material in RNA containing viruses like TMV (Tobacco Mosaic Virus) and they separated RNA from the protein of TMV viruses. Three molecular biologists in the early 1980's (Leslie Orgel, Francis Brick and Carl Woese) independently proposed the '**RNA world**' as the first stage in the evolution of life, a stage when RNA catalysed all molecules necessary for survival and replication. The term 'RNA world' first used by **Walter Gilbert** in 1986, hypothesizes RNA as the first genetic material on earth. There is now enough evidence to suggest that essential life processes (such as metabolism, translation, splicing etc.,) evolved around RNA. RNA has the ability to act as both genetic material and catalyst. There are several biochemical reactions in living systems that are catalysed by RNA. This catalytic RNA is known as **ribozyme.** But, RNA being a catalyst was reactive and hence unstable. This led to evolution of a more stable form of DNA, with certain chemical modifications. Since DNA is a double stranded molecule having complementary strand, it has resisted changes by evolving a process of repair. Some RNA molecules function as gene regulators by binding to DNA and affect gene expression. Some viruses use RNA as the genetic material. Andrew Fire and Craig Mellow (recipients of Nobel Prize in 2006) were of the opinion that RNA is an active ingredient in the chemistry of life. The types of RNA and their role have been discussed in class XI.

5.6 Properties of genetic material

The experiment by Hershey and Chase clearly indicates that it is DNA that acts as a genetic material. However, in some viruses like Tobacco mosaic virus (TMV), bacteriophage θB, RNA acts as the genetic material. A molecule that can act as a genetic material should have the following properties:

- **Self Replication:** It should be able to replicate. According to the rule of base pairing and complementarity, both nucleic acids (DNA and RNA) have the ability to

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